

Linear Algebra

Question difficulty is indicated by 🐣 (lower difficulty), 🐣🐣 (greater difficulty), and 🐣🐣🐣 (greatest difficulty). A topic label is also given on the question.

Linear Transformation 🐣

Problem 1

SMMC 2021 B1 Let $n \geq 2$ be an integer, and let $\mathbf{O}$ be the $n \times n$ matrix whose entries are all equal to 0.

Two distinct entries of the matrix are chosen uniformly at random, and those two entries

are changed from 0 to 1. Call the resulting matrix $\mathbf{A}$. Determine the probability that $\mathbf{A}^2 = \mathbf{O}$, as a function of n .

Linear Independence 🐣

Problem 2

Putnam 2003 Do there exist real polynomials $a(x), b(x), c(x), d(x)$ such that

$$1 + xy + (xy)^2 = a(x)c(y) + b(x)d(y)$$

holds identically?

Linear Equations 🐣🐣

Problem 3

Let $A = (a_{ij})$ be a real $n \times n$ matrix satisfying

$$|a_{ii}| > \sum_{j \neq i} |a_{ij}|$$

for all $1 \leq i \leq n$. Prove that A is invertible.

Linear Transformations and Eigenvector 🐣🐣

Problem 4

SMMC 2025 A2: Let p be a prime number. Given a square matrix M with entries in $\mathbb{Z}/p\mathbb{Z}$, an eigenvector of M is a non-zero vector $\vec{v}$ with entries in $\mathbb{Z}/p\mathbb{Z}$ such that $M\vec{v} = \lambda\vec{v}$ for some $\lambda \in \mathbb{Z}/p\mathbb{Z}$. For each positive integer n , determine the average number of eigenvectors of an $n \times n$ matrix with entries in $\mathbb{Z}/p\mathbb{Z}$. (Here $\mathbb{Z}/p\mathbb{Z}$ refers to the integers modulo p .)

Block Matrices and Determinant 🦊🦊🦊

Problem 5

Putnam 2018-A-2 Let $S_1, S_2, \dots, S_{2^{n-1}}$ be the nonempty subsets of $\{1, 2, \dots, n\}$ in some order, and let M be the $(2^{n-1}) \times (2^{n-1})$ matrix whose (i, j) entry is

$$m_{ij} = \begin{cases} 0 & \text{if } S_i \cap S_j = \emptyset; \\ 1 & \text{otherwise.} \end{cases}$$

Calculate the determinant of M .

Powers of Matrices and Nil-potency 🦊🦊🦊

Problem 6

Putnam 2012-A-5 Let $\mathbb{F}_p$ denote the field of integers modulo a prime p , and let n be a positive integer.

Let v be a field vector in $\mathbb{F}_p^n$ and let M be an $n \times n$ matrix with entries in $\mathbb{F}_p$, and define $G : \mathbb{F}_p^n \rightarrow \mathbb{F}_p^n$ by $G(x) = v + Mx$. Let G^k denote the k -fold composition of G with itself, that is $G^1(x) = G(x)$ and $G^{(k+1)}(x) = G(G^k(x))$.

Determine all pairs p, n for which there exist v and M such that the p^n vectors $G^k(0), k = 1, 2, \dots, p^n$ are distinct.